

Grid Connection Study:
50 MW Solar Farm – Victoria 132 KV Network

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Software: pandapower (Python)

1. Introduction

The increasing penetration of renewable energy generation, particularly utility-scale solar photovoltaic (PV) systems, presents significant technical challenges for the operation of existing electricity networks. Unlike conventional synchronous generation, inverter-based PV systems introduce unique characteristics including variable output, limited fault current contribution, and potential voltage rise at the point of connection. These characteristics must be carefully assessed prior to connection to ensure the stability, reliability and security of the existing network are maintained.

This report presents a grid connection feasibility study of a 50 MW solar photovoltaic farm connecting to a regional 132 kV transmission network in Victoria, Australia. The study follows the general principles of AEMO's generator connection process under the National Electricity Rules (NER).

The assessment covers steady-state load flow analysis under multiple operating conditions, voltage sensitivity analysis across the full generator output range, and three-phase fault level analysis. The study was conducted using pandapower, an open-source Python-based power systems analysis tool. Network parameters are based on typical values for Victorian transmission infrastructure, and all assumptions are documented in Section 3.

The consequences of connecting generation without adequate prior assessment can include voltage violations, thermal overloading of network equipment, and in severe cases, threats to system security and supply continuity. This study aims to identify any such impacts and confirm the technical feasibility of the proposed connection before any physical works are undertaken.

2. Network Description

2.1. Network Topology

The network model consists of a 132kV external grid representing the upstream Victorian transmission network with a maximum short circuit level of 2500 MVA. A 132kV overhead transmission line of 25km connects the grid to the high voltage side of a 60MVA 132/33kV zone substation transformer. The low voltage side of the transformer connects to a 33kV bus, from which a 2km 33kV collector feeder connects to the solar farm connection point. The proposed 50MW solar PV farm is modelled as a static generator operating at unity power factor.

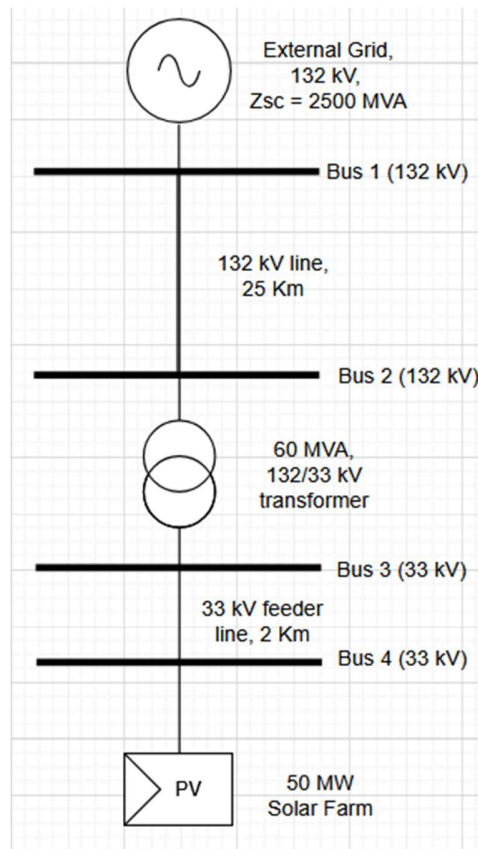


Figure 1: Single Line Diagram, 50 MW solar farm grid connection study

2.2. Network Parameters

Table 1: Network Element Parameters

Element	Details
External Grid	132 kV, $Z_{sc} = 2500$ MVA, $R/X = 0.1$
132 kV line	25 km, $r = 0.1 \Omega/\text{km}$, $x = 0.4 \Omega/\text{km}$, $I_{max} = 0.4$ kA
Transformer	60 MVA, 132/33 kV, $v_k = 12.5\%$, $v_{kr} = 0.1\%$
33kV Feeder	2 km, $r = 0.15 \Omega/\text{km}$, $x = 0.35 \Omega/\text{km}$, $I_{max} = 1.0$ kA
Solar Farm	50 MW, 33 kV, unity power factor, $k = 1.0$

- Parameters based on typical values for Victorian transmission infrastructure. Complete pandapower network model provided in Appendix A.

2.3. Assumptions

As actual network data for the Victorian transmission network is commercially sensitive and not publicly available, the following assumptions were made in developing the network model for this study:

- The upstream Victorian transmission network is represented as an external grid with a maximum short circuit level of 2500 MVA and minimum of 2000 MVA, consistent with typical values for a regional 132kV connection point.
- Transmission line parameters are based on typical values for ACSR overhead conductors used in Australian 132kV and 33kV networks.
- The 132/33kV transformer rating of 60 MVA is consistent with a typical Victorian zone substation transformer.
- The 33kV collector feeder is assumed to be a dedicated connection rated for the full 50MW solar farm output, with a thermal current limit of 1.0 kA.
- The solar farm is modelled at unity power factor for the base case, consistent with AEMO's default assumption for inverter-based generation.
- The solar inverter short circuit current ratio is assumed as $k = 1.0$, typical for utility-scale PV inverters.
- Voltage compliance limits of 0.90 to 1.10 p.u are applied at all buses in accordance with NER S5.1.4.

- Switchgear fault ratings are assumed as 31.5 kA for 132kV equipment and 25.0 kA for 33kV equipment, consistent with standard Australian switchgear ratings.

3. Methodology

3.1. Software

The power system analysis was conducted using pandapower, an open-source Python-based power systems simulation tool developed by Fraunhofer IEE. pandapower uses a Newton-Raphson load flow algorithm and implements the IEC 60909 standard for short circuit analysis. All simulations were performed in a Jupyter Notebook environment on a Windows platform.

3.2. Load Flow Analysis

Steady-state load flow analysis was performed to assess the impact of the solar farm on network voltages and equipment loading. The Newton-Raphson method was used for all load flow calculations. Two operating cases were studied, base case with no solar generation, representing pre-connection network conditions and full solar output of 50MW, representing the worst case voltage rise condition.

In addition, a sensitivity analysis was conducted by varying the solar farm active power output from 0MW to 50MW in 10MW increments to assess voltage behaviour across the full operating range.

3.3. Fault Analysis

Three-phase fault level analysis was conducted in accordance with IEC 60909 using pandapower's short-circuit module. Fault levels were calculated at all network buses under two conditions, with and without the solar farm connected, to quantify the fault current contribution of the proposed connection. The solar farm inverter was modelled with a short circuit current ratio of $k = 1.0$, consistent with typical inverter-based PV generation.

3.4. Compliance Criteria

The following compliance criteria were applied in assessing the feasibility of the proposed connection:

Table 2: Compliance Criteria

Criterion	Limit	Reference
Voltage (normal operation)	0.90 - 1.10 pu	NER S5.1.4
Line Thermal Loading	$\leq 100\%$	Equipment rating
Transformer loading	$\leq 100\%$	Equipment rating
Fault Current (132 kV)	≤ 31.5 kA	Assumed switchgear rating
Fault Current (33 kV)	≤ 25.0 kA	Assumed switchgear rating

4. Results

4.1. Load Flow Analysis

4.1.1. Case 1: Without Solar Connection

The base case load flow was performed with the solar farm disconnected from the network, representing pre-connection conditions. The results establish a voltage baseline against which the impact of the solar farm connection can be assessed. Bus voltage results are presented in Table 3.

Table 3: Bus Voltage Results, No Solar

Bus	Name	Voltage (p.u)	Angle (Degrees)
1	Grid Connection 132 kV	1	0
2	Substation HV 132 kV	1.000	-0.01
3	Substation LV 33 kV	1.000	-0.013
4	Solar Farm 33 kV	1.000	-0.013

All bus voltages are within the compliance range of 0.90 to 1.10 p.u. Voltages are close to nominal (1.0 pu) across all buses as expected with no generation connected. Line and transformer loadings are negligible in the base case.

4.1.2. Case 2: Full Solar Output

The full output case was performed with the solar farm operating at its rated output of 50MW, representing the worst-case voltage rise condition. Bus voltage and equipment loading results are presented in Table 4 and Table 5 respectively.

Table 4: Bus Voltage Results, Full Solar Output

Bus	Name	Voltage (p.u)	Angle (Degrees)
1	Grid Connection 132 kV	1	0
2	Substation HV 132 kV	1.007	1.682
3	Substation LV 33 kV	0.999	7.546
4	Solar Farm 33 kV	1.012	9.367

Table 5: Equipment Loading Results, Full Solar Output

Equipment	Loading (%)	Limit (%)
132 kV line	53.97	100
33 kV Feeder line	86.44	100
132/33 kV Transformer	82.34	100

Voltage rise is observed at the solar farm bus (Bus 4) compared to the base case, which is characteristic of embedded generation injecting active power into the network. All voltages remain within compliance limits. Equipment loading is within thermal limits across all network elements.

4.2. Sensitivity Analysis

A sensitivity analysis was conducted by varying the solar farm active power output from 0MW to 50MW in 10MW increments. The voltage at the solar farm bus (Bus 4) was recorded at each output level to assess voltage behaviour across the full operating range. Results are illustrated in Figure 2.

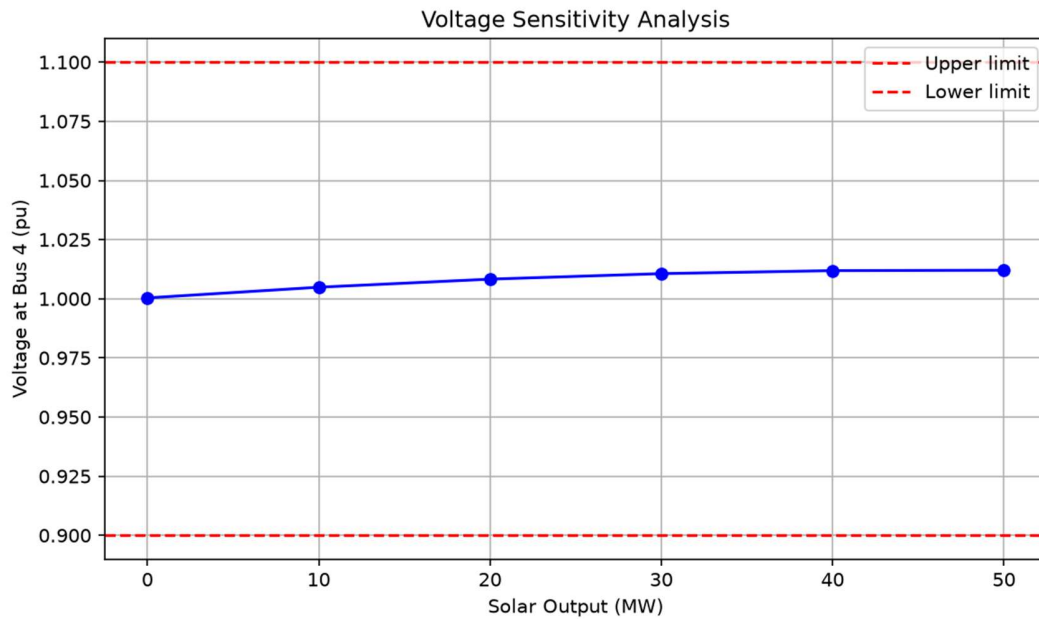


Figure 2: Variation of voltage with solar generation

Figure 2 shows a progressive voltage rise at Bus 4 as solar output increases, which is consistent with the expected behaviour of inverter-based generation. The voltage remains within the compliance limits of 0.90 to 1.10 p.u across the full output range, confirming that no voltage violations occur under any operating conditions studied.

4.3. Fault Analysis

Three-phase fault level analysis was conducted at all network buses with and without the solar farm connected. Results are presented in Table 6.

Table 6: Fault Level Results

Bus	Name	Without Solar (kA)	With Solar (kA)	Increase (kA)	Rating (kA)
1	Grid 132 kV	10.935	11.153	0.22	31.5
2	Substation HV 132 kV	4.607	4.826	0.22	31.5
3	Substation LV 33 kV	6.311	7.186	0.87	25
4	Solar Farm 33 kV	5.172	6.047	0.87	25

The solar farm contributes a modest increase in fault current of 0.22 kA at the 132kV buses and 0.87 kA at the 33kV buses. This limited fault current contribution is characteristic of inverter-based generation, which is current-limited by inverter controls unlike conventional synchronous generators. All fault levels remain well within the assumed switchgear ratings at every bus.

5. Feasibility Assessment

Based on the load flow and fault level analysis conducted in this study, the technical feasibility of the proposed 50MW solar farm connection to the Victorian 132kV network is assessed against the compliance criteria established in Section 3. The results of each criterion are summarised in Table 7.

Table 7: Feasibility Assessment Summary

Criterion	Limit	Result	Status
Voltage at Bus 1	0.90 - 1.10 p.u	1	Within Limits
Voltage at Bus 2	0.90 - 1.10 p.u	1.007	Within Limits
Voltage at Bus 3	0.90 - 1.10 p.u	0.999	Within Limits
Voltage at Bus 4	0.90 - 1.10 p.u	1.012	Within Limits
132 kV Line Loading	$\leq 100\%$	53.97	Within Limits
33 kV Feeder Line Loading	$\leq 100\%$	86.44	Within Limits
Transformer Loading	$\leq 100\%$	82.34	Within Limits
Fault Level at Bus 1	≤ 31.5 kA	11.153	Within Limits
Fault Level at Bus 2	≤ 31.5 kA	4.826	Within Limits
Fault Level at Bus 3	≤ 25.0 kA	7.186	Within Limits
Fault Level at Bus 4	≤ 25.0 kA	6.047	Within Limits

As demonstrated in Table 7, all technical compliance criteria are satisfied across the full range of operating conditions studied. Key findings are summarised below:

- Voltage compliance: Bus voltages remain within the permissible range of 0.90 to 1.10 p.u at all buses under all operating conditions, including full solar output of 50MW. A progressive voltage rise was observed at the solar farm bus (Bus 4) as solar output increased, which is characteristic of inverter-based embedded generation. No voltage violations were identified.
- Thermal loading: All network equipment including the 132kV transmission line, 33kV collector feeder and 132/33kV transformer remained within their respective thermal limits under full solar output. No equipment overloading was identified.
- Fault level: The solar farm contributes a modest increase in fault current due to the current-limited nature of inverter-based generation. All fault levels remain well within the assumed switchgear ratings of 31.5 kA at 132kV and 25.0 kA at 33kV. No switchgear upgrades are required because of this connection.

Based on the above assessment, the proposed 50MW solar farm connection to the Victorian 132kV network is considered technically feasible under the studied conditions.

It is noted that this study is based on assumed network parameters and a simplified network model. The following limitations apply:

- Actual network parameters would need to be obtained from the relevant Network Service Provider (AusNet Services) prior to formal connection application.
- Dynamic and transient stability studies, harmonic analysis, and protection coordination studies have not been conducted and would be required as part of a full connection assessment.
- The study does not account for N-1 contingency conditions, which would be required under AEMO's connection assessment process.
- Generator Performance Standard (GPS) compliance under NER clause 5.2 has not been formally assessed.

6. Conclusion

This study presented a grid connection feasibility assessment for a proposed 50MW solar photovoltaic farm connecting to a regional 132kV transmission network in Victoria, Australia. The assessment was conducted using pandapower, an open-source Python-based power systems analysis tool, and covered steady-state load flow analysis, voltage sensitivity analysis, and three-phase fault level analysis.

The key findings of the study are as follows:

- Voltage compliance was maintained at all network buses across the full solar output range of 0 to 50MW. A progressive voltage rise was observed at the solar farm connection point (Bus 4) as solar output increased, which is consistent with the expected behaviour of inverter-based embedded generation. The maximum voltage recorded was approximately 1.012 p.u at Bus 4 under full solar output, within the permissible limit of 1.10 p.u.
- All network equipment remained within thermal limits under full solar output. The 132kV transmission line, 33kV collector feeder and 132/33kV transformer recorded maximum loadings of 53.97%, 86.44% and 82.34% respectively, all below the 100% thermal limit.
- The solar farm contributed a modest increase in fault current of 0.22 kA at the 132kV connection point and 0.87 kA at the 33kV solar farm bus, consistent with the current-limited characteristics of inverter-based generation. All fault levels remained well within assumed switchgear ratings.
- Based on the above findings, the proposed 50MW solar farm connection is considered technically feasible at the assumed connection point under the operating conditions studied.

It is recommended that the following additional studies be conducted prior to formal connection application to AEMO:

- Dynamic and transient stability studies to assess generator performance under network disturbances
- Harmonic analysis to assess the impact of inverter switching on network power quality
- Protection coordination study to ensure correct fault detection and clearance
- N-1 contingency analysis to assess network performance under single element outage conditions
- Generator Performance Standard (GPS) compliance assessment under NER clause 5.2

This study demonstrates the application of power systems analysis techniques to a realistic renewable energy grid connection scenario, using industry-standard methodologies and compliance criteria consistent with AEMO's connection assessment process.

References

- Australian Energy Market Operator (AEMO), National Electricity Rules, Schedule 5.1, clause S5.1.4, 2024.
- IEC 60909-0:2016, Short-circuit currents in three-phase AC systems.
- Glover, J.D., Sarma, M.S. and Overbye, T.J., Power Systems Analysis and Design, 6th ed., Cengage Learning, 2017.
- pandapower Development Team, pandapower Documentation, Fraunhofer IEE, 2024. Available at: www.pandapower.org

Appendix

Pandapower Code:

```
from pandapower.create import (  
    create_empty_network,  
    create_ext_grid,  
    create_transformer_from_parameters,  
    create_bus,  
    create_switch,  
    create_sgen,  
    create_line_from_parameters)  
from pandapower.run import runpp  
import matplotlib.pyplot as plt  
import pandapower.plotting as plot  
from pandapower.shortcircuit import calc_sc  
  
# Building the network with the PV system connected.  
net = create_empty_network()  
bus1 = create_bus(net, name = "Bus 1", vn_kv = 132)  
bus2 = create_bus(net, name = "Bus 2", vn_kv = 132)  
bus3 = create_bus(net, name = "Bus 3", vn_kv = 33)  
bus4 = create_bus(net, name = "Bus 4", vn_kv = 33)  
grid = create_ext_grid(net, bus1, vm_pu = 1.0, s_sc_max_mva = 2500, s_sc_min_mva =  
2000, rx_max = 0.1, rx_min = 0.1)  
line1 = create_line_from_parameters(net, from_bus = bus1, to_bus = bus2, length_km = 25,  
r_ohm_per_km = 0.15, x_ohm_per_km = 0.4, c_nf_per_km = 10, max_i_ka = 0.4, name =  
"132 KV line")
```

```
trafo1 = create_transformer_from_parameters(net, hv_bus = bus2, lv_bus = bus3, sn_mva = 60, vn_hv_kv = 132, vn_lv_kv = 33, vkr_percent = 0.1, vk_percent = 12.5, pfe_kw = 50, i0_percent = 0.1, name = "Main Transformer 132/33 KV")
```

```
line2 = create_line_from_parameters(net, from_bus = bus3, to_bus = bus4, length_km = 2, r_ohm_per_km = 0.15, x_ohm_per_km = 0.35, c_nf_per_km = 10, max_i_ka = 1.0, name = "Substation connection line")
```

```
solar = create_sgen(net, bus = bus4, p_mw = 50, q_mvar = 0, sn_mva = 50, k = 1.0, name = "50 MW Solar Plant", type = "PV", scaling = 1.0)
```

```
# With no solar connection
```

```
net.sgen.p_mw = 0
```

```
runpp(net)
```

```
case_1_bus = net.res_bus.copy()
```

```
case_1_line = net.res_line.copy()
```

```
case_1_trafo = net.res_trafo.copy()
```

```
print("Case 1: No solar")
```

```
print("\n Bus Voltages")
```

```
print(case_1_bus[['vm_pu','va_degree']])
```

```
print("\n Line Loadings")
```

```
print(case_1_line[['loading_percent']])
```

```
print("\n Transformer Loading")
```

```
print(case_1_trafo[['loading_percent']])
```

```
# With Solar at full output
```

```
net.sgen.p_mw = 50
```

```
runpp(net)
```

```
case_2_bus = net.res_bus.copy()
```



```

case_2_line = net.res_line.copy()
case_2_trafo = net.res_trafo.copy()
print("Case 2: Solar at full output")
print("\n Bus Voltages")
print(case_2_bus[['vm_pu','va_degree']])
print("\n Line Loadings")
print(case_2_line[['loading_percent']])
print("\n Transformer Loading")
print(case_2_trafo[['loading_percent']])

# Sensitivity Analysis
output_levels = [0,10,20,30,40,50]
voltages = []
for p in output_levels:
    net.sgen.p_mw = p
    runpp(net)
    voltages.append(net.res_bus.vm_pu.iloc[-1])
plt.figure(figsize=(9,5))
plt.plot(output_levels, voltages, marker='o', color='blue')
plt.axhline(y=1.10, color='r', linestyle='--', label='Upper limit')
plt.axhline(y=0.90, color='r', linestyle='--', label='Lower limit')
plt.xlabel("Solar Output (MW)")
plt.ylabel("Voltage at Bus 4 (pu)")
plt.title("Voltage Sensitivity Analysis")
plt.legend()
plt.grid(True)

```

```
plt.savefig("voltage_profile.png", dpi=150)
```

```
plt.show()
```

```
# Fault Analysis
```

```
# Without Solar
```

```
net.sgen.in_service = False
```

```
calc_sc(net, fault = "3ph", case = "max")
```

```
fault_without_solar = net.res_bus_sc.copy()
```

```
print("Fault without Solar:")
```

```
print(fault_without_solar[['ikss_ka','skss_mw']])
```

```
# With Solar
```

```
net.sgen.in_service = True
```

```
calc_sc(net, fault = "3ph", case = "max")
```

```
fault_with_solar = net.res_bus_sc.copy()
```

```
print("\n Fault with Solar")
```

```
print(fault_with_solar[['ikss_ka','skss_mw']])
```